Evaluation of clusters in DNA methylation array data of head neck squamous cell carcinoma

Spring 2022 STAT 6500 Project

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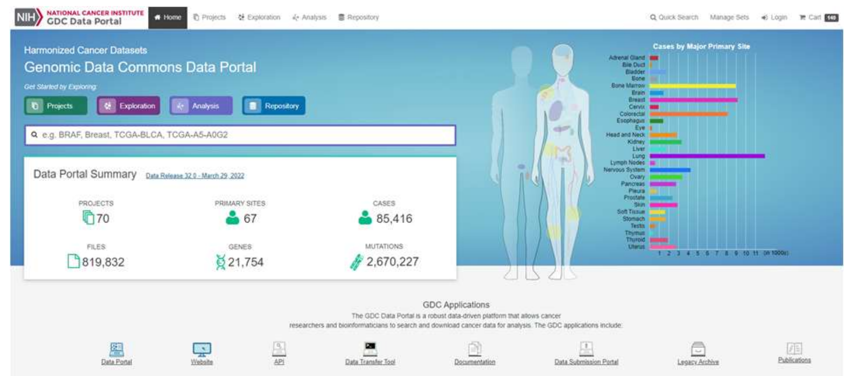
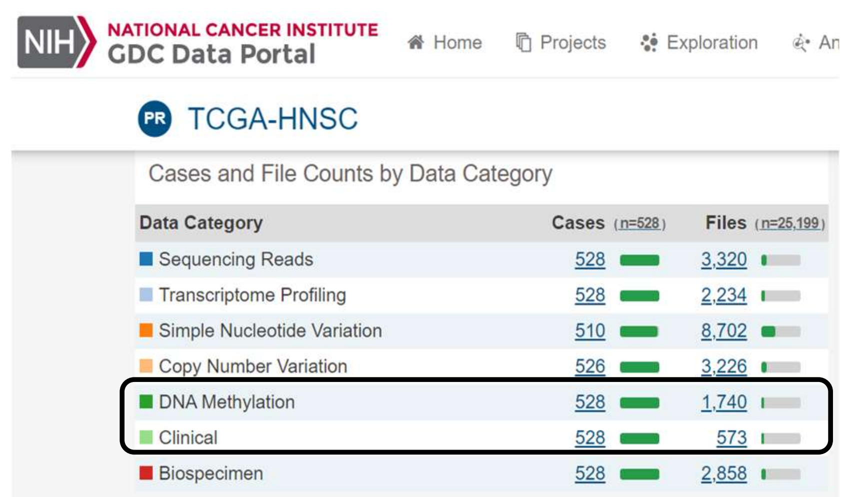
Dataset: TCGA-HNSC

Characteristic		N(%)
Median age at diagnosis in year (range)		62(20-90)
Gender	Male	386 (73%)
	Female	142 (27%)
Race	White	452 (86%)
	Black	48 (9%)
	Asian and others	13 (2.5%)
Vital status	Alive	358 (68%)
	Dead	170 (32%)
Tumor status	Tumor free	339 (64%)
	With tumor	139 (36%)
……		





portal.gdc.cancer.gov

Main questions

- Can patients be grouped by methylation alone?
- Are these groupings associated with survival?
- A multivariate model to predict the patients' survival outcomes?

Methylation array specifics

- Illumina's Infinium HumanMethylation450 BeadChip
  - Assays 450,000+ methylation sites, coded as "cg" followed by a number
- **C**ytosine-**p**hosphate-**G**uanine (CpG) sites are enriched for methylation
- Matrix of CpGs can be called **beta values** due to its distribution

Clustering methods used

- K-means
- Partitioning around medoids (PAM)
- Spectral clustering

- Tried a range of K from 2 to 6

K-means

- stats::kmeans()
- Finds clusters s.t. total within-cluster variation is minimized
- "Within-cluster variation" can be measured in various ways
  - Default: Euclidean distance
  - The distance is computed between observations and a **centroid**
- 10 random initializations were used

An Introduction to Statistical Learning, 2nd Ed.

PAM

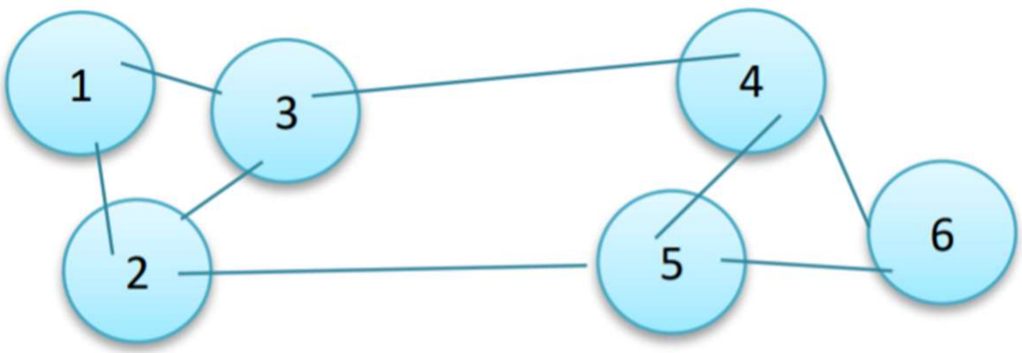
- cluster::pam()
- Minimize the average dissimilarity of observations to the **medoid**, which is an observation located centrally in the cluster

https://www.cs.umb.edu/cs738/pam1.pdf

Spectral clustering

- kernlab::specc()

Example of a "graph" and its matrix representation



	1	2	3	4	5	6
1	0	1	0	0	1	1
2	1	0	1	0	1	0
3	0	1	0	1	0	0
4	1	0	1	0	1	1
5	0	1	0	1	0	1
6	0	1	1	0	1	0

https://www.mygreatlearning.com/blog/introduction-to-spectral-clustering/

Spectral clustering

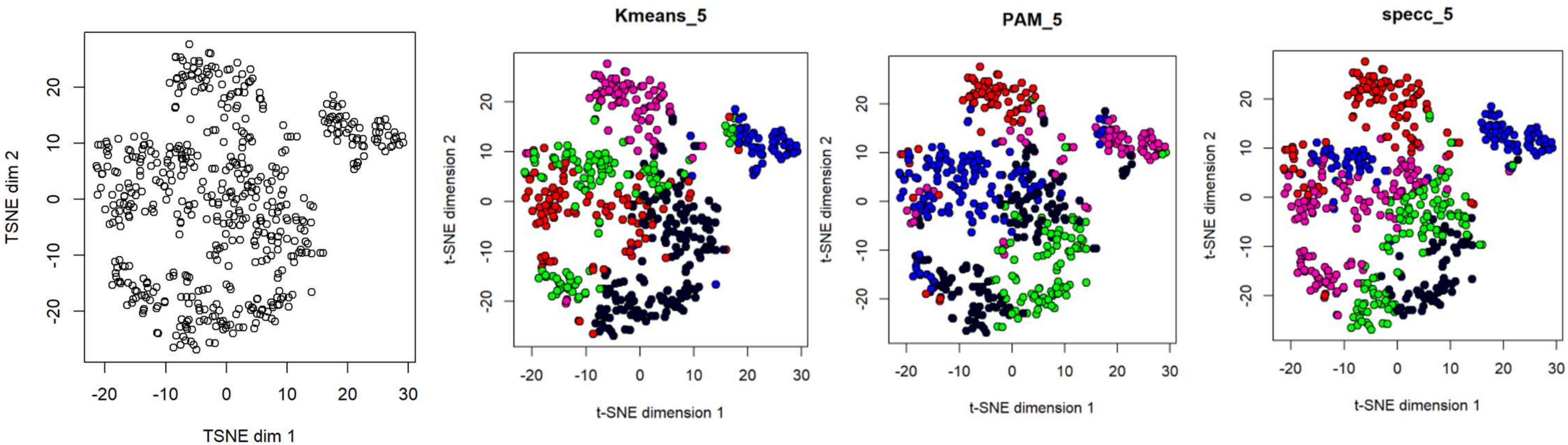
Gaussian similarity function

$$K(\mathbf{x}, \mathbf{x}') = \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|^2}{2\sigma^2}\right)$$

1) Create affinity matrix "W" where edges are weighted by K(x, x')
2) Create diagonal degree matrix "D" by taking the sum of the number of edges connected to the node
3) Create Laplacian "L" where L = D – W
4) Spectral decomposition of L to get eigenvalues
5) Collect the eigenvectors $u_1 \ldots u_k$ as columns of a matrix U
6) Use k-means clustering on matrix U to generate clusters

A Tutorial on Spectral Clustering by von Luxburg, 2007

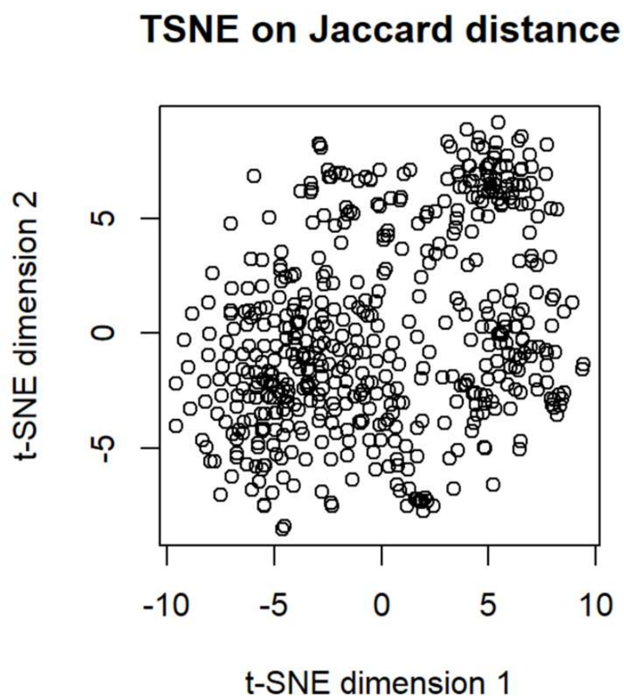
When using the unaltered CpG matrix for clustering, the clusters appear dubious

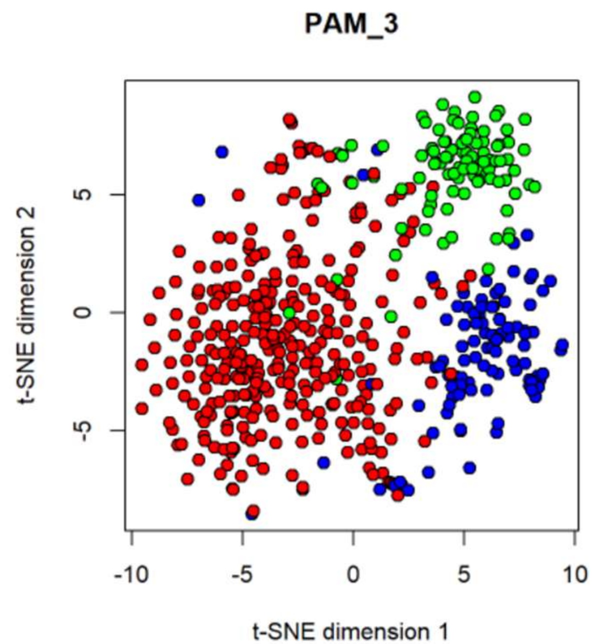


How about using the Jaccard distance for clustering?

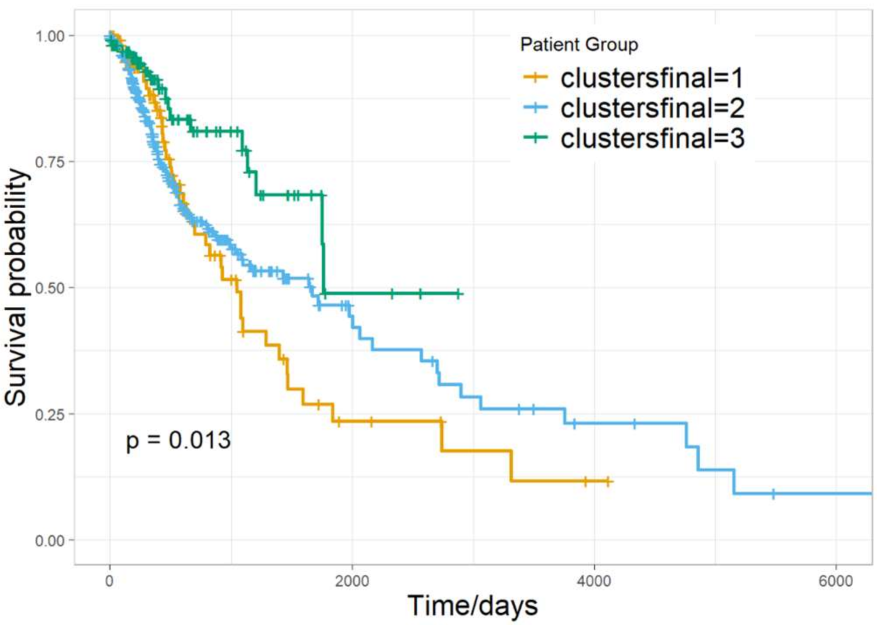
- Mercator::binaryDistance(cpgmatrix, metric="jaccard")
- Measures dissimilarity between sample sets
- Beta-distributed values is close enough to binary to generate a useful binary distance matrix
- Feed binary distance matrix into TSNE and clustering algorithms

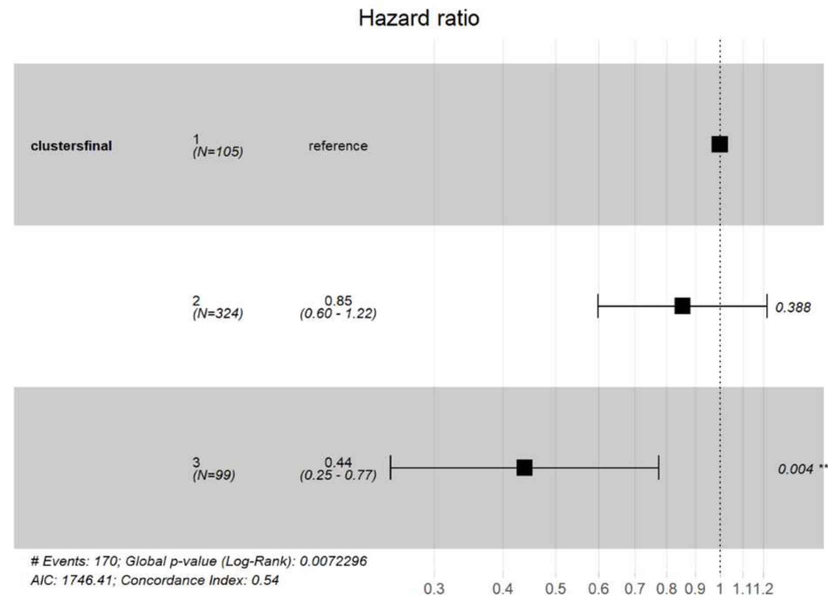
When using the Jaccard distance for TSNE, 3 putative clusters appear, which was identified by PAM k=3





PAM k=3 clustering outcomes are associated with patient survival

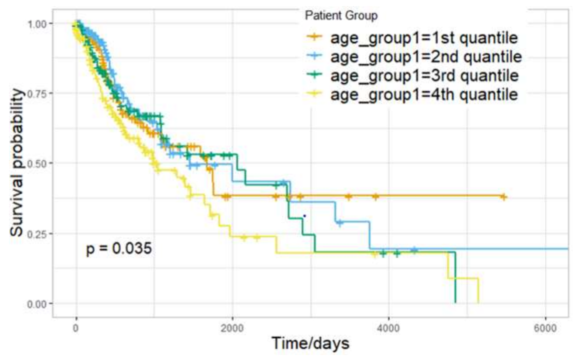


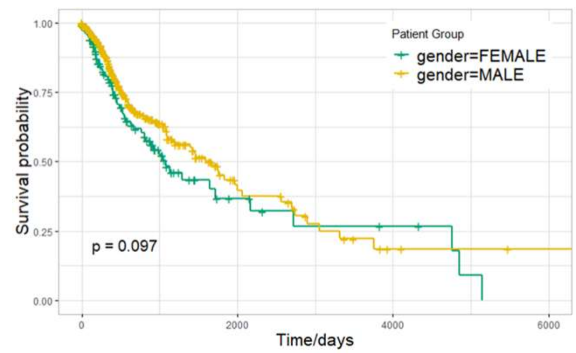


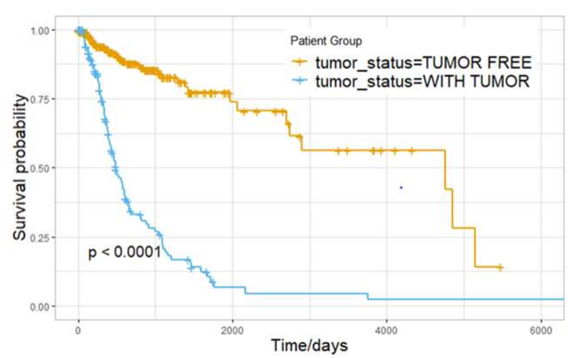
Other clinical features vs. survival outcomes

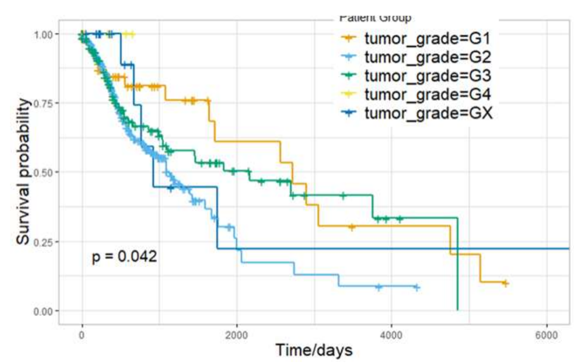
Clinical features checked:

- Gender
- Age in years
- Tumor status
- Tumor grade
- ~~Race or ethnicity~~
- ~~History of other malignancy~~
- ~~AJCC staging information~~



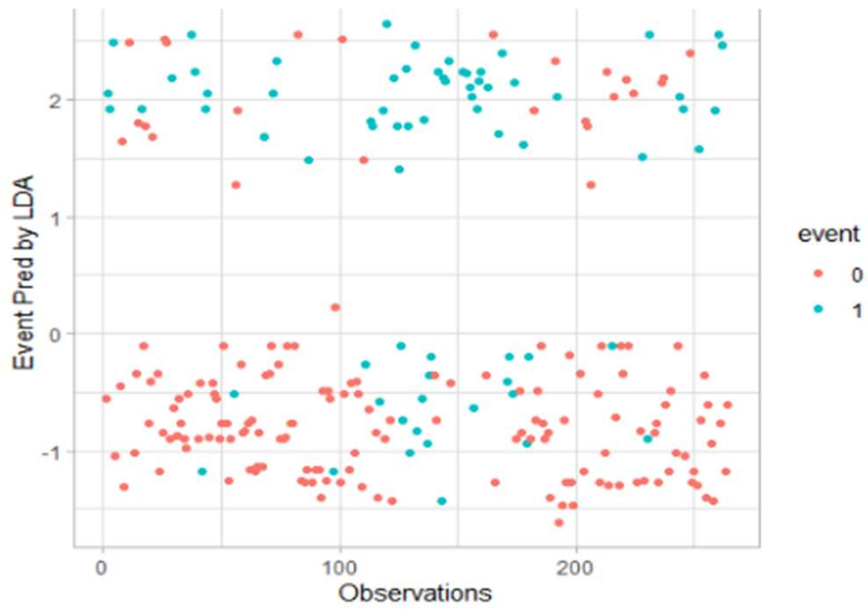






A multivariate model to predict the clinical survival outcomes

- Randomly split the samples into training and test (50% and 50%) set
- Select the variables that are associated with survival in univariate analysis
- **Linear discriminant analysis** (LDA)
- Test error rate: ~0.2




```
lda.fit = lda(event ~ gender +age_group1 + tumor_status + tumor_grade + MethyGroup,
              data = clin_dat_train)
```

```
lda.pred = predict(lda.fit, clin_dat_test)
```

Conclusions

- Can patients be grouped by methylation alone?

>> When using the Jaccard distance for TSNE on methylation data, 3 putative clusters appear, which was identified by PAM k=3.

- Are these groupings associated with survival?

>> Methylation clustering outcomes are associated with survival; the association is as strong as other clinical annotations such as age group and tumor grade.

- A multivariate model to predict the patients' survival outcomes?

>>The performance of simple LDA model, using methylation clustering, gender, age group, tumor status and tumor grade as input variables, to predict the clinical survival outcomes needs to be further improved.

>> The prognostic value of methylation data in HNSCC study